

Name: _____

Student ID: _____

If (the last 3 digits of your student ID) mod 3 = m and ((the last 3 digits of your student ID) mod 5 = n , then you need to solve the m -th problem from the first group and the n -th problem from the second group.

a) Design Turing machines for the following languages using the simulator provided (for manual and more details, see <https://schaetztc.github.io/tursi/>): (10)

0. $\{a^n b^n c^n \mid n \geq 0\}$
1. $\{a^n b^m c^n d^m \mid n, m \geq 1\}$
2. $\{w \in \{a, b, c\}^* \mid \#a(w) = \#b(w) = \#c(w)\}$ i.e. the set of all strings with equal number of a , b and c .

b) Design Turing machines to perform the following operations (the leftmost bit is LSB) using the simulator provided (for manual and more details, see <https://schaetztc.github.io/tursi/>) (10)

0. Increment (add 1). The tape initially contains wc , where w is a binary number. Your TM should add 1 to w and write the sum to the right of the c .
1. Decrement (subtract 1). The tape initially contains wc , where w is a binary number. Your TM should subtract 1 from w and write the result to the right of the c .
2. Logical XOR. The tape initially contains $w_1 c w_2 c$, where w_1 and w_2 are binary numbers. Your TM should compute logical XOR of w_1 and w_2 and write the result to the right of the second c .
3. Logical AND. The tape initially contains $w_1 c w_2 c$, where w_1 and w_2 are binary numbers. Your TM should compute logical AND of w_1 and w_2 and write the result to the right of the second c .
4. Logical OR. The tape initially contains $w_1 c w_2 c$, where w_1 and w_2 are binary numbers. Your TM should compute logical OR of w_1 and w_2 and write the result to the right of the second c .

Submission instructions:

Submit the *.tm files in a zipped folder, named using your student ID, through moodle.

Deadline: 03/04/2026 (Friday) 11:59pm